DONI VISHWA VARDHAN

AI/ML Engineer

Email | +91 8639685756 | github | LinkedIn | Portfolio

ACADEMIC DETAILS

Indian Institute of Technology Delhi *2026*

B.Tech in Biochemical Engineering & Biotechnology — CGPA: 7

EXPERIENCE

Management Trainee – Program Manager | Basic Enterprises *June 2026 – Present*

Products: CleverTap • Agent Flash • Invoice Rejection Tracker • Agent Retention Strategies

- Designed, built, and deployed a Retrieval-Augmented Generation (RAG) AI agent for Agent Support and Call Center operations, taking it from prototype to internal production use.
- Led end-to-end product development, from Figma wireframes and requirement scoping to feature delivery, using data-driven insights to solve user problems.
- Analyzed CleverTap product analytics to identify user drop-offs and pain points, prioritized the roadmap, and collaborated with engineering and operations teams to define success metrics and deliver data-informed solutions.

PROJECTS

Support Triage MCP Server — M8ven Verified Publisher |

Model Context Protocol (MCP), FAISS, Python,FastAPI

- Built an **MCP server** and a **FastAPI** web dashboard over the same triage tools (classify, draft reply, duplicate check, file and update cases, queue Q&A), where filing a case runs triage live. Added **rate limiting**, **input validation** and **HTML escaping** for a public demo.
- Grounded **gpt-oss-20b (Groq)** classification in similar resolved tickets. On 200 held-out Banking77 queries grouped into 9 **routing** queues, **accuracy** rose from **79.0% zero-shot to 93.5%** (95% CI 89.2 to 96.2), against 68.5% for retrieval alone. Built **duplicate detection** whose F1 rose from **0.71 (hashing) to 0.82 (BGE-small embeddings)** on 24 labeled pairs.
- Wrote 120+ pytest tests with GitHub Actions CI and an evaluation harness that reports confidence intervals.

Financial Intelligence Agent | GitHub

Python, LangGraph, Qdrant, PostgreSQL, FastAPI, Docker, Ollama

- Built a RAG-based financial research agent(Parsing, Data Cleaning, Chunking, Embedding) to analyze annual reports and financial documents using open-source LLMs (via **Ollama**), **LangGraph**, and **Qdrant** vector store with hybrid **dense+BM25** retrieval.
- Developed tool-using agent workflows (PostgreSQL **text-to-SQL**, financial calculations, document retrieval) exposed as APIs via FastAPI, with plan-then-execute orchestration and **sqlglot**-validated SQL execution.
- Implemented an **LLM-as-judge** evaluation harness scoring **answer correctness = 4.4/5**, **groundedness = 4.6/5**, and **routing accuracy = 94.6%**, with a defense-in-depth SQL safety layer (validated + read-only) and Dockerized deployment.

Loan Default Risk Modeling (Home Credit) | GitHub

Python, pandas, scikit-learn, LightGBM, SHAP, MLflow, FastAPI

- Built a loan-default model on Home Credit data (307K loans, 6 tables), engineering 176 features from credit-bureau, installment and card histories to lift CV **ROC-AUC from 0.757 to 0.779.**
- Final **LightGBM** scores **0.78 ROC-AUC / 0.28 PR-AUC** (3.5x base rate) on a held-out test set; a cost-based cut-off rejects 51% of defaulters and 17% of applicants (+22% profit vs approving all, assumed costs).
- Served it via FastAPI with **SHAP**-based risk factors, tracked runs in **MLflow**, added PSI drift monitoring and 11 pytest tests; dropped the gender feature after showing it cost under 0.002 ROC-AUC.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, Dart

ML / Deep Learning: Scikit-learn, PyTorch, TensorFlow, Keras, XGBoost, Hugging Face Transformers, NLP, Numpy, Pandas

GenAI & LLM Engineering: LangChain, LangGraph, RAG pipelines, Prompt Engineering, Vector DBs (Chroma, FAISS, Qdrant), Ollama, Fine-tuning (LoRA/QLoRA, PEFT, trl, bitsandbytes), Guardrails (Llama Guard), Gateways (LiteLLM), Model Context Protocol (MCP)

MLOps & Deployment: FastAPI, Docker, Kubernetes, MLflow, Azure ML, AWS SageMaker, Vertex AI, CI/CD (GitHub Actions), Data Bricks, React, Vercel, Next JS, REST APIs

Cloud: Microsoft Azure, AWS,GCP

Tools: Git, VS Code, MS Excel, MS Office